2025-Winter Paper 2 – Urology

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Q1. Describe etiology, clinical presentation and recent advances in management of Urethral Structure Disease

Type: LAQ (25 marks) (also answered in 2016-Summer Paper 2, Q2)

"Structure" is taken to mean urethral stricture disease.

Definition

- A fixed narrowing of the **anterior** urethra (inside the corpus spongiosum) that will not accept instruments without tearing the mucosa. It is an epithelial scar, often with **spongiofibrosis**.
- Terminology (WHO (World Health Organization) 2004, ICUD (International Consultation on Urological Diseases) 2010): after pelvic fracture, **PFUI (pelvic fracture urethral injury)** or distraction defect; after prostate surgery or radiation, **stenosis or contracture**.
- Halving calibre (30 to 15 Fr) loses about three-quarters of lumen area, so symptoms appear late.
- **Anatomy**: bulbar urethra is eccentric (dorsal) in the spongiosum. Dual blood supply (bulbar arteries plus retrograde flow from the glans) lets the bulb be divided and mobilised.

Etiology

Category	Examples and mechanism
Idiopathic (commonest, about equal to iatrogenic)	No cause found
Iatrogenic	Catheters (very common in LMICs (low- and middle-income countries)), TURP (transurethral resection of prostate), TURBT (transurethral resection of bladder tumour), cystoscopy, hypospadias repair, radiotherapy. Mechanism: mucosal injury, pressure ischaemia, infection
Traumatic	**Straddle injury**: bulb crushed against the pubis; pelvic fracture gives a posterior obliterative defect
Inflammatory	Gonorrhoea (historically commonest, now rare); **lichen sclerosus (LS)**. Chlamydia: no proven link
Congenital	Rarest; short anterior stricture in infants

- **LS** starts at the glans and meatus. High-pressure voiding forces urine into Littre glands, causing inflammation and deep spongiofibrosis. It **recurs in any genital skin** used for

repair.
- **Jordan spectrum**: mucosal fold → iris constriction → full-thickness → deep spongiofibrosis → fibrosis beyond spongiosum → fistula or abscess.

Clinical presentation

- Poor stream, straining, spraying, incomplete emptying, post-void dribble; UTI (urinary tract infection), prostatitis, epididymitis.
- **Retention** after long-tolerated symptoms; periurethral abscess, fistula ("watering-can perineum"), stones, upper-tract damage.

Evaluation: define location, length, depth, density

- **Uroflowmetry** (flat plateau curve) and PVR (post-void residual).
- **Dynamic RUG (retrograde urethrogram) + VCUG (voiding cystourethrogram)**, done by the surgeon, oblique views.
- **Flexible urethroscopy**; **sonourethrography** for spongiofibrosis; MRI (magnetic resonance imaging) for PFUI only.
- In retention: **suprapubic catheter**, not blind dilation. **Urethral rest** (6–8 weeks) shows the true length.

Standard management (brief)

- First agree the aim: **palliation or cure**.
- **Dilation** (balloon over a guidewire) stretches, not tears; cures only epithelial strictures.
- **DVIU (direct vision internal urethrotomy)**: heals by secondary intention, a race between epithelialisation and wound contraction. Only for **bulbar, <1.5 cm, no dense spongiofibrosis, first attempt**. A second DVIU adds almost nothing. Laser equals cold knife.
- **Urethroplasty**:

Technique	Use
EPA (excision and primary anastomosis)	**Gold standard**; short bulbar (1–2 cm, sometimes 3–5 cm proximally). Excise all scar, spatulate, tension-free
BMG (buccal mucosal graft) onlay	Longer bulbar: dorsal (Barbagli), ventral, dorsal inlay (Asopa). Graft equals flap
Penile skin flaps (Orandi, circular)	Penile strictures (not LS)
Staged repair (Johanson)	LS, failed hypospadias, infection
Perineal urethrostomy	Panurethral disease, elderly, multiple failures

- Onlay beats tube repairs. **PFUI**: suprapubic catheter, then delayed **perineal anastomotic repair** with progressive steps (crural separation, inferior pubectomy, rerouting).

Recent advances in management

1. **Ladder abandoned**: the "reconstructive ladder" is archaic. Repeated endoscopy lowers later urethroplasty success, so **early urethroplasty** is offered after one failed DVIU (AUA (American Urological Association) and EAU (European Association of Urology) guidelines).
2. **Optilume paclitaxel-coated balloon** (FDA (Food and Drug Administration) 2023): paclitaxel inhibits fibroblasts. **ROBUST III**: stricture-free 74.6% vs 26.8% at 1 year after failed endoscopy; IPSS (International Prostate Symptom Score) 9 vs 19.9.
3. **Anti-scarring adjuncts after DVIU**: mitomycin C injection reduces recurrence; intermittent self-dilation; removable stents. Permanent UroLume stents withdrawn.

4. **Vessel-sparing and non-transecting EPA**: preserve bulbar arteries, theoretically less ED (erectile dysfunction).
5. **Augmented anastomosis**: excise the narrowest segment, graft the rest.
6. **Kulkarni one-sided dorsal BMG** through a perineal incision for panurethral strictures.
7. **Staged oral mucosa repair for LS**; avoid skin. **BMG perineal urethrostomy** reduces neomeatal stenosis.
8. **Robotic reconstruction** of bladder neck contracture and posterior stenosis (Y-V plasty, BMG).
9. **Tissue-engineered oral mucosa** (investigational). **Prevention**: silicone and hydrophilic catheters, smaller endoscopes.
10. **PROMs (patient-reported outcome measures)**: USS-PROM (Urethral Stricture Surgery PROM) is increasingly used to judge success.

- **Also know:** dartos-island fossa repair; lingual graft; clobetasol for early LS.

Complications and follow-up

- Recurrence (EPA: mostly in first 6 months; grafts decline with time), ED, chordee, dribble, graft-site pain. Follow up with flow, PVR and symptom score.

Diagram to draw: RUG of a bulbar stricture, and cross-section of dorsal versus ventral onlay.

> **Key points to remember:**
> - Idiopathic and iatrogenic are commonest; LS recurs in skin.
> - Define location, length, depth, density (dynamic RUG + VCUG).
> - DVIU only once, bulbar <1.5 cm; EPA is gold standard.
> - Optilume (ROBUST III 74.6% vs 26.8%); early urethroplasty, ladder abandoned.

Q2. Describe use and principles of immunology and Immunotherapy in Urological Oncology

Type: LAQ (25 marks) (immunomodulation in RCC (renal cell carcinoma): see 2025-Winter Paper 4, Q3a)

Introduction

- Immunotherapy works by either **supplying a danger signal** that starts a Th1 response (BCG (bacille Calmette-Guérin), vaccine adjuvant) or **removing a brake** on an existing response (PD-1 (programmed death-1), CTLA-4 (cytotoxic T-lymphocyte antigen-4)).

Principles of tumour immunology

Arm	Cells and role
Innate (rapid, pattern-based)	PAMPs (pathogen-associated molecular patterns) bind **TLRs (Toll-like receptors)**; macrophages, neutrophils; urothelium as sentinel (IFN (interferon), IL-8 (interleukin-8))
Bridge	**Dendritic cell (DC)**: takes up antigen, matures on danger signal, migrates to lymph node
Adaptive (specific, memory)	**CD8 killer** (sees class I MHC (major histocompatibility complex), ~9 amino acids); **CD4 helper** (class II); B cells
Brake	**Treg** (regulatory T cell, FoxP3+ CD4); MDSCs (myeloid-derived suppressor cells); M2 macrophages

- **Two signals**: TCR (T-cell receptor) meets MHC–peptide (signal 1) plus **CD28–B7** costimulation (signal 2).
- **Th1 vs Th2**: IL-12 gives Th1 (IL-2, IFN-γ, TNF-α (tumour necrosis factor-α)), which is antitumour. IL-4 gives Th2 (antibody, chronic inflammation).
- **Checkpoints**: CTLA-4 (on Tregs) competes for B7; PD-1 on CD8 cells binds **PD-L1** (programmed death ligand-1); LAG-3 binds class II.
- **Adaptive immune resistance**: CD8 cells secrete IFN-γ, which induces PD-L1 on the tumour, which switches the same T cells off. So PD-L1 marks a suppressed ongoing response.

Immunoediting (three Es)

- **Elimination** (small tumours destroyed) → **Equilibrium** (growth held) → **Escape** (clinical cancer).
- Escape: loss of antigen or MHC, Treg expansion, checkpoint ligands.
- **GU (genitourinary) landscape**: bladder: inflammation-driven (schistosomiasis), CD8 infiltration good. Kidney: **CD8 paradox** (more CD8, worse outcome). Prostate: Treg-dominated, PD-L1 low, so checkpoint inhibitors fail.

Immunotherapy in practice

1. BCG in NMIBC (non-muscle-invasive bladder cancer)

- Bacilli attach via **fibronectin**; TLR2/4 activation → IL-8 and TNF-α → neutrophil wave → DC priming → **Th1 CD4** response → memory.
- Th1 predicts success, Th2 failure. About half stay in remission for years. Used for intermediate/high-risk NMIBC and CIS (carcinoma in situ) with induction plus maintenance.

2. Cytokines and gene therapy

- **IFN-α** and **high-dose IL-2** (durable complete responses in a few mRCC (metastatic renal cell carcinoma) patients; capillary leak). Intravesical IFN-α: mixed.

- **Nadofaragene firadenovec** (adenovirus with IFN-α2b gene + Syn3): BCG-unresponsive CIS, CR (complete response) 53.4%; FDA (Food and Drug Administration) 2022.
- **N-803** (IL-15 superagonist) with BCG: FDA approved 2024 for BCG-unresponsive CIS.

3. Cancer vaccines

- Antigen plus adjuvant (often GM-CSF (granulocyte-macrophage colony-stimulating factor)) to activate DCs.
- **Sipuleucel-T**: autologous blood cells pulsed with **PAP (prostatic acid phosphatase)–GM-CSF**, three infusions. Asymptomatic mCRPC (metastatic castration-resistant prostate cancer): **OS (overall survival) 25.8 vs 21.7 months**, **no delay in progression**. First FDA-approved solid-tumour vaccine.
- Failed: PROSTVAC (prostate), IMA901 and AGS-003 (RCC).

4. Checkpoint inhibitors

Site	Use (trial)
BCG-unresponsive CIS	Pembrolizumab if cystectomy declined/unfit (KEYNOTE-057: CR 41%)
Adjuvant MIBC (muscle-invasive bladder cancer) and UTUC (upper tract urothelial carcinoma)	Nivolumab after radical surgery (**CheckMate 274**: DFS (disease-free survival) 20.8 vs 10.8 months; included UTUC, benefit uncertain in that small subgroup). Atezolizumab (IMvigor010) negative
Metastatic UC (urothelial carcinoma)	Avelumab **maintenance** after platinum (JAVELIN Bladder 100, OS gain); pembrolizumab second line (**KEYNOTE-045**: OS 10.3 vs 7.4 months); **enfortumab vedotin (EV) + pembrolizumab**
RCC	Nivolumab + ipilimumab (CheckMate 214); IO (immuno-oncology) + TKI (tyrosine kinase inhibitor) combinations; adjuvant pembrolizumab (KEYNOTE-564, DFS HR (hazard ratio) 0.68)
Prostate	Monotherapy poor; pembrolizumab only for **MSI-H/dMMR** (microsatellite instability-high/mismatch repair-deficient) tumours

- VEGF (vascular endothelial growth factor) suppresses DCs and T-cell entry, so TKIs synergise with IO.
- **Biomarkers**: PD-L1 CPS (combined positive score) only for first-line single-agent pembrolizumab in cisplatin-ineligible UC; TMB (tumour mutational burden), MSI.

5. Adoptive cell therapy (brief)

- **CAR-T (chimeric antigen receptor T cells)**: T cells engineered against a surface antigen (e.g. PSMA (prostate-specific membrane antigen)). Investigational in GU; limited by poor trafficking, hostile microenvironment and cytokine release syndrome.

Toxicity: irAEs (immune-related adverse events)

- Colitis, hepatitis, pneumonitis, thyroiditis, hypophysitis, nephritis, myocarditis, rash.

Grade	Action
1	Continue, monitor
2	Hold; prednisolone 0.5–1 mg/kg
3–4	Stop; methylprednisolone 1–2 mg/kg; infliximab or mycophenolate if refractory

- Endocrinopathies need hormone replacement. BCG toxicity: cystitis, BCG sepsis.

Recent advances

- **EV-302**: enfortumab vedotin + pembrolizumab first line in metastatic UC, OS 31.5 vs 16.1 months versus platinum; now the preferred first line.

- **NIAGARA**: perioperative durvalumab with neoadjuvant cisplatin chemotherapy improved event-free and overall survival in MIBC.
- **Also know:** CheckMate 901 (nivolumab + gemcitabine–cisplatin); ADT (androgen deprivation) + IO trials; anti-CD38 and LAG-3 targets.

Diagram to draw: T cell–tumour synapse with TCR–MHC, CD28–B7, CTLA-4 and PD-1/PD-L1 blocked by antibodies.

Key points to remember:

- Three Es: elimination, equilibrium, escape; Th1 is antitumour.
- BCG: TLR2/4 → neutrophils → Th1 memory.
- Sipuleucel-T: OS +4.1 months, no delay in progression.
- CheckMate 274 adjuvant nivolumab; EV-302 EV + pembrolizumab first line.

Q3a. Non obstructive azoospermia

Type: SAQ (10 marks) (also answered in 2019-Summer Paper 1, Q3d)

Definition

- **NOA (non-obstructive azoospermia):** no sperm in the **centrifuged** ejaculate on two samples, due to **failure of spermatogenesis**. About 60% of azoospermic men.

Causes

- **Genetic: Klinefelter (47,XXY)**, commonest; **Y microdeletion** (AZF (azoospermia factor) a, b, c).
- **Acquired:** cryptorchidism, mumps orchitis, torsion, **chemotherapy or radiation**, trauma, varicocele.
- **Idiopathic**: commonest overall.
- *Note:* hypogonadotropic hypogonadism is pre-testicular and treated medically (hCG (human chorionic gonadotropin) ± FSH (follicle-stimulating hormone)).

Evaluation

- Small soft testes, palpable vasa, normal ejaculate volume.
- **FSH >7.6 IU/L and testis long axis ≤4.6 cm** suggest NOA.
- LH (luteinising hormone), testosterone, oestradiol, prolactin.
- **Karyotype and Y microdeletion in all** (AUA/ASRM (American Urological Association/American Society for Reproductive Medicine)), with counselling.

Histology	Meaning	Sperm retrieval
Hypospermatogenesis	All stages, fewer cells	Best
Maturation arrest	Halts at one germ-cell stage	About 40%
SCO (Sertoli cell–only)	No germ cells	About 30%

Management

1. Optimisation: stop exogenous testosterone; repair clinical varicocele (evidence weak); off-label clomiphene, anastrozole or hCG for low testosterone.

2. Micro-TESE (microdissection testicular sperm extraction)

- **Principle:** spermatogenesis in NOA is **focal**. At ×20–25 magnification, **larger, opaque tubules** are taken; little tissue removed, blood supply spared.
- Sperm found in **about 50%**; better than conventional TESE.

Predictor	Retrieval (%)
Post-orchidopexy	74
AZFc deletion	70
Klinefelter	68
Post-chemotherapy	53
AZFa or AZFb complete deletion	**0: do not operate**

- FSH and testis size do not reliably predict retrieval.

3. ICSI (intracytoplasmic sperm injection) with fresh or frozen sperm. Sons of AZF deletion carriers inherit the deletion.

4. No sperm: donor sperm or adoption.

Also know: in-vitro spermatogenesis and stem-cell transplantation are experimental.

Diagram to draw: opened testis under microscope: plump tubules versus thin Sertoli cell–only tubules.

Key points to remember:

- FSH >7.6 and testis ≤4.6 cm suggest NOA.
- Karyotype and Y microdeletion in all.
- Micro-TESE: sperm in about half; SCO 30%.
- AZFa/b complete deletion: no retrieval.

Q3b. Retroileal Ureter

Type: SAQ (10 marks)

"Retroileal" is taken as retroiliac ureter, the nearest real entity; retrocaval ureter is compared below (full answer: 2017-Winter Paper 3, Q3d).

Definition

- The ureter passes **behind the common iliac artery** and is compressed there.
- Better called "**preureteral iliac artery**": the anomaly is presumed vascular; the ureter is trapped.

Embryology

- Normally the **ventral root of the umbilical artery** is replaced by a more **dorsal** branch, which lies behind the ureter.
- If the **ventral root persists** and the dorsal root fails, the ureter is trapped behind the artery.

Features

- **Very rare**; either side; **bilateral** cases reported.
- Obstruction at **L5–S1**.
- **Associated anomalies are common**, especially **vasal and mesonephric duct ectopia**: ureter and vas draining together into the posterior urethra; vasa ending in the ureters; ectopic vaginal ureter.

Presentation and diagnosis

- Flank pain, UTI (urinary tract infection) or hydroureteronephrosis down to the iliac crossing.
- **CT urography (computed tomography urography) with arterial phase** or MRI (magnetic resonance imaging) shows ureter and artery together; **diuretic renogram** for function and drainage.
- Caution: a vessel indenting an already dilated ureter may not be the cause.

Management

- Non-obstructed: observe.
- Obstructed: **divide the ureter, re-route it in front of the artery** and do a spatulated ureteroureterostomy (or reimplantation) over a double-J stent; open, laparoscopic or robotic.

Comparison with retrocaval ureter

	Retroiliac	Retrocaval (circumcaval)
Vessel	Common iliac artery	IVC (inferior vena cava)
Embryology	Persistent ventral umbilical artery root	Persistent subcardinal vein
Side	Either, may be bilateral	Right
Level	L5–S1	L3–L4, "fishhook"
Associations	Vasal/mesonephric ectopia	Horseshoe kidney, left renal anomalies

- **Also know:** distal ureter obstruction by uterine, umbilical, obturator or hypogastric vessels.

Diagram to draw: ureter behind the common iliac artery at L5–S1, dilated above.

Key points to remember:

- Retroiliac = preureteral iliac artery; persistent ventral umbilical artery root.
- Obstruction at L5–S1; either side or bilateral.
- Look for vasal and ureteric ectopia.
- Retrocaval: right, L3–L4, fishhook, 1 in 1500.

Q3c. Vascular access for hemodialysis

Type: SAQ (10 marks) (also answered in 2020-Summer Paper 4, Q3b)

Introduction

- Haemodialysis needs blood flow of **250–350 mL/min**. Access failure is a major cause of morbidity in ESKD (end-stage kidney disease).

Types

Access	Construction	Ready	Comment
Radiocephalic AVF (arteriovenous fistula; Brescia–Cimino)	Radial artery to cephalic vein at wrist, end-to-side	6–8 weeks	**First choice**: best patency, fewest infections
Brachiocephalic AVF	At elbow	6–8 weeks	Higher flow, more steal
AVG (arteriovenous graft)	PTFE (polytetrafluoroethylene) loop or straight	2–3 weeks	More stenosis, thrombosis, infection
Tunnelled cuffed CVC (central venous catheter)	Right internal jugular	At once	Bridge or last resort; infection, central stenosis

- Non-tunnelled catheters: acute dialysis only.

Planning

- **Hierarchy**: AVF, then AVG, then CVC; **distal before proximal; non-dominant arm first**. KDOQI (Kidney Disease Outcomes Quality Initiative) 2019 favours an individual "ESKD Life-Plan".
- **Timing**: refer at eGFR (estimated glomerular filtration rate) <30; create the AVF about **6 months** before dialysis.
- **Vein preservation**: no venepuncture or PICC (peripherally inserted central catheter) in that arm; avoid subclavian lines.
- **Assessment**: pulses, **Allen test** (ulnar collateral flow), **Doppler vein mapping** (vessels about ≥2 mm).
- **Maturity "rule of 6s"**: flow >600 mL/min, diameter >6 mm, depth <6 mm, at 6 weeks.

Complications

Complication	Treatment
Failure to mature (juxta-anastomotic stenosis)	Angioplasty, ligate side branches
Stenosis (neointimal hyperplasia)	PTA (percutaneous transluminal angioplasty), stent graft
Thrombosis (usually on a stenosis)	Thrombectomy, then fix the stenosis
Steal syndrome (hand ischaemia)	Banding, DRIL (distal revascularisation–interval ligation)
Aneurysm; infection (CVC > AVG > AVF)	Repair; antibiotics, excise graft

- **Also know:** brachiobasilic transposition AVF; venous hypertension; high-output cardiac failure.

Monitoring

- Thrill and bruit, arm-elevation test, prolonged bleeding after needling, rising venous pressure; duplex when abnormal.

Recent advances

- **Endovascular AVF** (percutaneous creation); drug-coated balloons; bioengineered vessels.

Diagram to draw: radiocephalic end-to-side fistula at the wrist.

> **Key points to remember:**
> - AVF first, distal, non-dominant arm, 6 months ahead.
> - Rule of 6s: >600 mL/min, >6 mm, <6 mm deep, 6 weeks.
> - Thrombosis sits on a stenosis; steal → banding or DRIL.

Q3d. Botulinum Toxin and its role in Urology

Type: SAQ (10 marks) (also answered in 2017-Winter Paper 3, Q3b)

Introduction

- *Clostridium botulinum* neurotoxin, serotypes A–G. Urology uses **onabotulinumtoxinA (BoNT-A)**.
- 150 kDa: heavy chain (100 kDa) plus zinc-dependent light chain (50 kDa).

Mechanism

1. Heavy chain binds the cholinergic nerve terminal; toxin is internalised.
2. Light chain enters cytosol and **cleaves SNAP-25** (synaptosomal-associated protein 25), a SNARE vesicle-docking protein. Vesicles cannot fuse, so **acetylcholine release stops**.

- **Afferent action**: less ATP (adenosine triphosphate), substance P and CGRP (calcitonin gene-related peptide); fewer TRPV1 (capsaicin) and P2X3 (purinergic) receptors on C fibres, so urgency falls.
- Effect wanes as terminals sprout (6–9 months).

Technique

- Cystoscopic detrusor injection, **20 sites (100 U) or 30 sites (200 U)**, sparing trigone; check PVR (post-void residual) at 2 weeks.

Urological uses

Indication	Dose and evidence
Idiopathic OAB (overactive bladder), third line	**100 U** (EMBARK). **ABC**: equal to antimuscarinic, more complete dryness. **ROSETTA**: 200 U comparable to SNM (sacral neuromodulation), more UTI (urinary tract infection)
NDO (neurogenic detrusor overactivity): SCI (spinal cord injury), MS (multiple sclerosis)	**200 U** (DIGNITY): less incontinence, larger capacity, lower detrusor pressure
Paediatric NDO (spina bifida)	About **10 U/kg (maximum 360 U)** in Campbell series; FDA (Food and Drug Administration) label (age ≥5) 6 U/kg, maximum 200 U
DSD (detrusor–sphincter dyssynergia)	Intrasphincteric injection
IC/BPS (interstitial cystitis/bladder pain syndrome)	100 U, often trigonal; AUA (American Urological Association) option, off-label
BPH (benign prostatic hyperplasia)	Intraprostatic: placebo-controlled RCTs (randomised controlled trials) showed **no benefit**; not recommended

- **Also know:** pelvic floor spasm.

Complications

- **Retention needing CISC (clean intermittent self-catheterisation)**, more in NDO; each 10 mL of baseline PVR raises risk about 27%. **Patient must accept CISC beforehand.**
- **UTI**; haematuria; rarely generalised weakness. Contraindications: active UTI, myasthenia gravis, pregnancy.

Diagram to draw: cholinergic synapse showing SNAP-25 cleavage by the light chain.

Key points to remember:

- Cleaves SNAP-25; efferent and afferent action.
- 100 U idiopathic OAB; 200 U NDO.
- Children: 10 U/kg (max 360 U) in series; FDA label 6 U/kg (max 200 U).
- Main risks: retention needing CISC, UTI.

Q3e. Tubeless PCNL

Type: SAQ (10 marks) (also answered in 2016-Summer Paper 4, Q3e)

Definition

- PCNL (percutaneous nephrolithotomy) that ends **without a nephrostomy tube**.
- **Tubeless**: drainage by a ureteral stent (double-J, or externalised stent or string). **Totally tubeless**: no stent and no tube.
- **Wickham (1984)** did the first tubeless PCNL; **Bellman (1997)** popularised it.

Rationale

- A large nephrostomy tube causes pain, more analgesia, longer stay and higher cost. Most tracts stop bleeding without tamponade.

Selection

Suitable	Keep a nephrostomy
Complete clearance, single tract	Pyonephrosis or infected stone
Minimal bleeding	Significant bleeding
Intact collecting system	Significant collecting-system perforation
Supracostal access, bilateral cases (shown safe)	Residual stone needing a second look

Technique (exit strategy)

1. Confirm clearance by rigid and **flexible nephroscopy** and fluoroscopy.
2. Antegrade double-J stent, or externalised stent or string.
3. Withdraw the sheath under vision; **safety wire is removed last**.
4. Watch the tract; apply pressure (rib as backstop in thin patients). Fibrin glue did not reduce bleeding in randomised trials; some use haemostatic sealants or tract cautery.
5. Urethral catheter overnight.

Evidence

- Meta-analyses of randomised trials: shorter stay and operative time.
- Less pain than large tubes (**>16 Fr**), but **no benefit over small 8–9 Fr tubes**.
- No difference in fever, infection, transfusion or stone-free rate.
- Stent discomfort: worse WiSQoL (Wisconsin Stone Quality of Life) scores than nephrostomy in two trials. Externalised stents come out before discharge without cystoscopy.
- Children: insufficient evidence.

Disadvantages and complications

- Loss of tract for a second look.
- Stent symptoms and a further procedure for removal.
- Urine leak or urinoma.
- Bleeding without tamponade: re-insert a balloon or Kaye catheter, then **angioembolisation**.

Current position

- Choose tubeless, small tube or stent case by case (AUA (American Urological Association)). Pairs well with mini-PCNL.

Diagram to draw: kidney with sealed tract and double-J stent.

Key points to remember:

- Tubeless = stent only; totally tubeless = neither.
- Benefit is mainly over large (>16 Fr) tubes.
- Keep a tube for bleeding, perforation, infection or second look.

Q3f. Surgery For Recurrant Vesico Vaginal Fistulae

Type: SAQ (10 marks) (timing of VVF repair: see 2021-Summer Paper 2, Q3b)

Introduction

- Recurrent VVF (vesicovaginal fistula): persists after a failed repair. **The first repair has the best chance**; redo tissue is scarred and poorly vascularised.
- Failure causes: tension, ischaemia, overlapping suture lines, infection, poor drainage, **malignancy, radiation**, missed second tract.

Evaluation before redo

- **Biopsy the tract** if any history of cancer (fistula may signal recurrence).
- **Cystoscopy and vaginoscopy**: size, number, relation to ureters and bladder neck.
- **CT urography (computed tomography urography)** to exclude a **ureterovaginal fistula**; dye test.
- Urodynamics for capacity and compliance; treat UTI (urinary tract infection), optimise nutrition.

Timing

- Wait until inflammation settles (commonly about 3 months). Radiation: **≥12 months** after radiotherapy; failed radiation repairs were redone after **≥6 months**.

Choice of route

Vaginal	Abdominal (open, laparoscopic, robotic)
Distal or trigonal, accessible	**Recurrent ≥2 cm**, supratrigonal, narrow vagina
Failed **Latzko** can be cured by a second Latzko	Ureter involved, needs reimplantation
Flap-splitting, offset flaps	Augmentation or **Tanagho bladder flap** needed
Less pain, shorter stay	**O'Conor** bivalving finds the scarred plane

- Robotic repair of recurrences gives high closure.

Operative principles

- Excise scar; wide mobilisation; tension-free, multilayer, **non-overlapping** closure.
- **Interposition flap is widely used in recurrent fistula** (accepted though not proven superior; an RCT showed no benefit in simple VVF):

Flap	Route	Use
Martius (labial fat pad, pudendal vessels)	Vaginal	Mid–distal, bladder neck; not apex
Peritoneal	Vaginal or abdominal	Apical post-hysterectomy
Omentum (J-flap)	Abdominal	Radiation, recurrent, large
Gracilis, Singapore flaps	Vaginal/perineal	Refractory, large tissue loss

- Ureteric stents if within 1 cm; fill test 200–300 mL; catheter **≥10–14 days**, then cystogram.

When repair is unwise

- Multiple failures, severe radiation damage, massive tissue loss: **conduit or continent diversion**; if unfit, ureteric occlusion with nephrostomies.
- **Also know:** rectus abdominis flap (last resort); more than one repair predicts residual incontinence.

Diagram to draw: Martius flap between bladder and vaginal closures.

Key points to remember:

- First repair has the highest success.
- Biopsy; exclude ureteric fistula before redo.
- Interposition flap (Martius, peritoneum, omentum) in recurrent repair.
- Diversion after multiple failures or severe radiation.

Rapid Revision Sheet – 2025-Winter Paper 2

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Describe etiology, clinical presentation and recent advances in management of Urethral Structure Disease (LAQ)

- Idiopathic and iatrogenic are commonest; LS recurs in skin.
- Define location, length, depth, density (dynamic RUG + VCUG).
- DVIU only once, bulbar <1.5 cm; EPA is gold standard.
- Optilume (ROBUST III 74.6% vs 26.8%); early urethroplasty, ladder abandoned.

Q2. Describe use and principles of immunology and Immunotherapy in Urological Oncology (LAQ)

- Three Es: elimination, equilibrium, escape; Th1 is antitumour.
- BCG: TLR2/4 → neutrophils → Th1 memory.
- Sipuleucel-T: OS +4.1 months, no delay in progression.
- CheckMate 274 adjuvant nivolumab; EV-302 EV + pembrolizumab first line.

Q3a. Non obstructive azoospermia (SAQ)

- FSH >7.6 and testis ≤4.6 cm suggest NOA.
- Karyotype and Y microdeletion in all.
- Micro-TESE: sperm in about half; SCO 30%.
- AZFa/b complete deletion: no retrieval.

Q3b. Retroileal Ureter (SAQ)

- Retroiliac = preureteral iliac artery; persistent ventral umbilical artery root.
- Obstruction at L5–S1; either side or bilateral.
- Look for vasal and ureteric ectopia.
- Retrocaval: right, L3–L4, fishhook, 1 in 1500.

Q3c. Vascular access for hemodialysis (SAQ)

- AVF first, distal, non-dominant arm, 6 months ahead.
- Rule of 6s: >600 mL/min, >6 mm, <6 mm deep, 6 weeks.
- Thrombosis sits on a stenosis; steal → banding or DRIL.

Q3d. Botulinum Toxin and its role in Urology (SAQ)

- Cleaves SNAP-25; efferent and afferent action.
- 100 U idiopathic OAB; 200 U NDO.
- Children: 10 U/kg (max 360 U) in series; FDA label 6 U/kg (max 200 U).
- Main risks: retention needing CISC, UTI.

Q3e. Tubeless PCNL (SAQ)

- Tubeless = stent only; totally tubeless = neither.
- Benefit is mainly over large (>16 Fr) tubes.
- Keep a tube for bleeding, perforation, infection or second look.

Q3f. Surgery For Recurrant Vesico Vaginal Fistulae (SAQ)

- First repair has the highest success.
- Biopsy; exclude ureteric fistula before redo.
- Interposition flap (Martius, peritoneum, omentum) in recurrent repair.
- Diversion after multiple failures or severe radiation.